

Create complete college-level documentation for a Bus Reservation – Seat Booking System based on the following Feature Set I:

Feature Set I

1. Passenger details
2. Bus details
3. Route details
4. Seat selection
5. Booking

Prepare the following four sections in simple, student-friendly language suitable for a college project report.

1. Requirements

Include:

- Introduction
- Objective of the system
- Functional requirements
- Non-functional requirements
- Hardware requirements
- Software requirements
- User requirements
- System constraints

Explain each point clearly and briefly according to the Bus Reservation – Seat Booking System.

2. Algorithm

Create a simple step-by-step algorithm for the system.

Use this sequence:

Start

↓

Enter passenger details

↓

Enter bus details
↓
Enter route details
↓
Display available seats
↓
Select seat
↓
Enter booking details
↓
Confirm booking
↓
Generate booking confirmation
↓
End

Use simple language and make the algorithm suitable for a college practical/project report.

3. Flowchart

Create a clear and standard flowchart for the Bus Reservation – Seat Booking System.

Use standard flowchart symbols:

- Oval → Start / End
- Parallelogram → Input / Output
- Rectangle → Process
- Diamond → Decision

Flowchart sequence:

Start
↓
Enter passenger details
↓
Enter bus details
↓
Enter route details
↓
Display available seats
↓

Select seat

↓

Is the selected seat available?

→ No → Display "Seat not available" → Select another seat

→ Yes → Confirm seat selection

↓

Enter booking details

↓

Confirm booking

↓

Generate booking confirmation

↓

End

Make the flowchart neat, properly aligned, easy to understand, and suitable for a college project.

4. ER Diagram

Create a clear Entity Relationship (ER) Diagram for the Bus Reservation – Seat Booking System.

Use these entities and attributes:

Passenger

- Passenger_ID (PK)
- Name
- Age
- Gender
- Phone

Bus

- Bus_ID (PK)
- Bus_Name
- Bus_Number
- Total_Seats

Route

- Route_ID (PK)

- Source
- Destination
- Departure_Time
- Arrival_Time

Seat

- Seat_ID (PK)
- Seat_Number
- Seat_Status
- Bus_ID (FK)

Booking

- Booking_ID (PK)
- Passenger_ID (FK)
- Bus_ID (FK)
- Route_ID (FK)
- Seat_ID (FK)
- Booking_Date

Show:

- Primary Keys (PK)
- Foreign Keys (FK)
- Entity names
- Attributes
- Relationships
- Cardinality such as 1:1 and 1:M
- Clearly labeled relationship lines

Suggested relationships:

- Passenger → Booking = 1:M
- Bus → Booking = 1:M
- Route → Booking = 1:M
- Bus → Seat = 1:M
- Seat → Booking = 1:M

Make the ER diagram simple, clean, properly connected, and suitable for a college project report.

